

HOG + Random Forest

Bottle crack patch classification report

Run: 20261003_155012

Labels: good = 0; crack = 1.

Existing splits were retained. Exact duplicate image pixels across splits were checked. Physical-bottle separation must already be correct in the folders.

Split	Good images	Crack images
train	315	70
valid	105	14
test	32	8

Model and selection

Image resize: (128, 128); grayscale.

HOG settings: {'orientations': 9, 'pixels_per_cell': (8, 8), 'cells_per_block': (2, 2), 'block_norm': 'L2-Hys', 'transform_sqrt': True, 'feature_vector': True}

Features per image: 8100.

No feature scaling was used. Random Forest was fitted on train only. Eight configurations were compared using validation crack F1.

Selected settings: {'n_estimators': 200, 'max_depth': None, 'min_samples_leaf': 1, 'max_features': 'sqrt', 'class_weight': 'balanced'}

Selected threshold: 0.29

The model remains fitted on train only. Training scores are resubstitution scores, validation scores are selection scores, and test scores are the held-out evaluation.

Evaluation metrics

Metric	Train	Valid	Test
Accuracy	1.0000	1.0000	0.9000
Balanced accuracy	1.0000	1.0000	0.8438
Crack precision	1.0000	1.0000	0.7500
Crack recall	1.0000	1.0000	0.7500
Crack F1	1.0000	1.0000	0.7500
Specificity	1.0000	1.0000	0.9375
ROC AUC	1.0000	1.0000	0.9727
Average precision (AP)	1.0000	1.0000	0.8979

Split	Log loss	TP	TN	FP	FN
train	0.0608	70	315	0	0
valid	0.1384	14	105	0	0
test	0.2803	6	30	2	2

Average precision summarizes the precision-recall curve; it is not trapezoidal PR area.

These are patch-classification results. Whole-bottle inspection requires a separate scanning and aggregation evaluation.

Runtime and model size

Total tuning time: 32.31 seconds.

Mean patch pipeline time: 57.77 ms over 40 test images.

Timing includes image reading, resizing, HOG extraction and classification, with warm-up excluded. Drive access and hardware affect these measurements.

Model size: 0.61 MB.

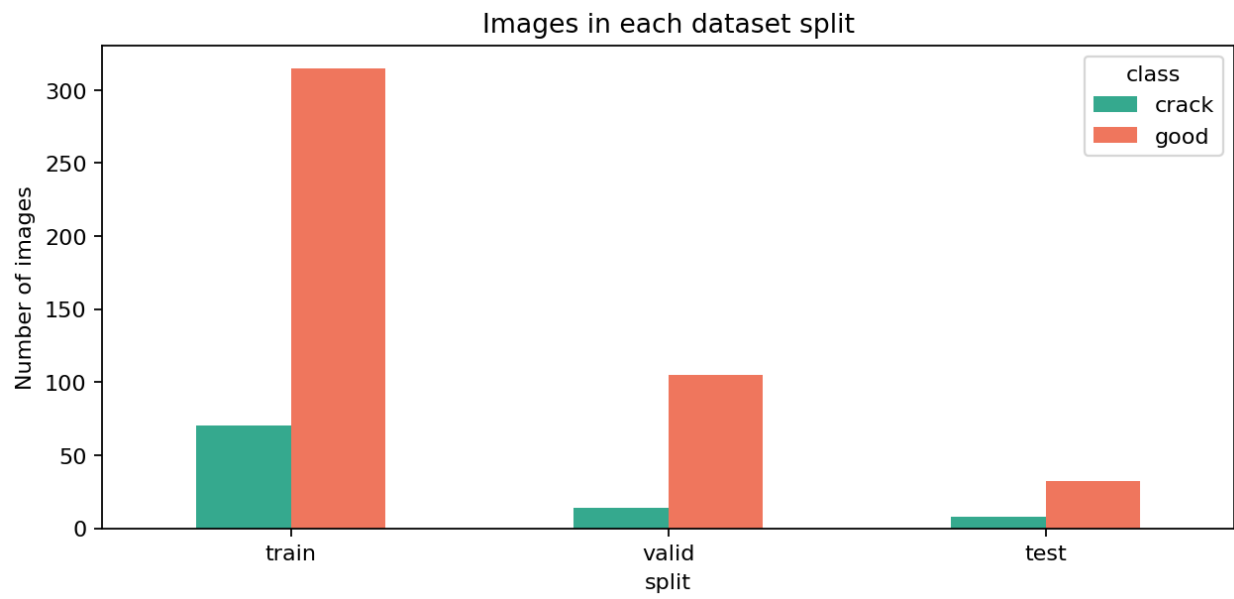
Python: 3.13.15; scikit-learn: 1.6.1; scikit-image: 0.25.2; Matplotlib: 3.10.0.

Feature importances are impurity-based diagnostics, not proof of which image regions caused a prediction.

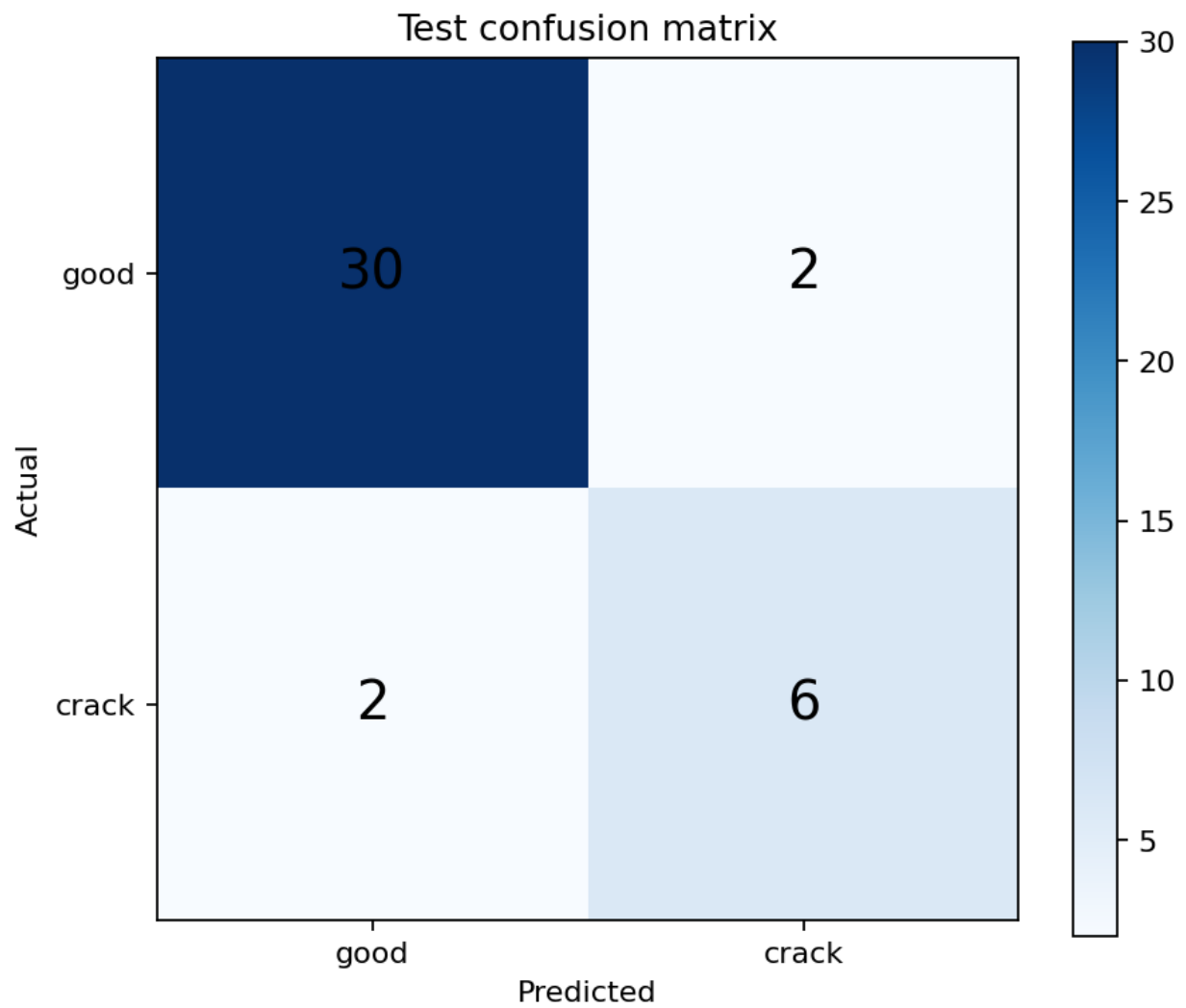
Validation search

Depth	Min leaf	Weight	Threshold	Valid F1
None	1	None	0.35	0.9333
None	1	balanced	0.29	1.0000
None	3	None	0.33	0.9333
None	3	balanced	0.35	1.0000
20	1	None	0.35	0.9333
20	1	balanced	0.29	1.0000
20	3	None	0.33	0.9333
20	3	balanced	0.35	1.0000

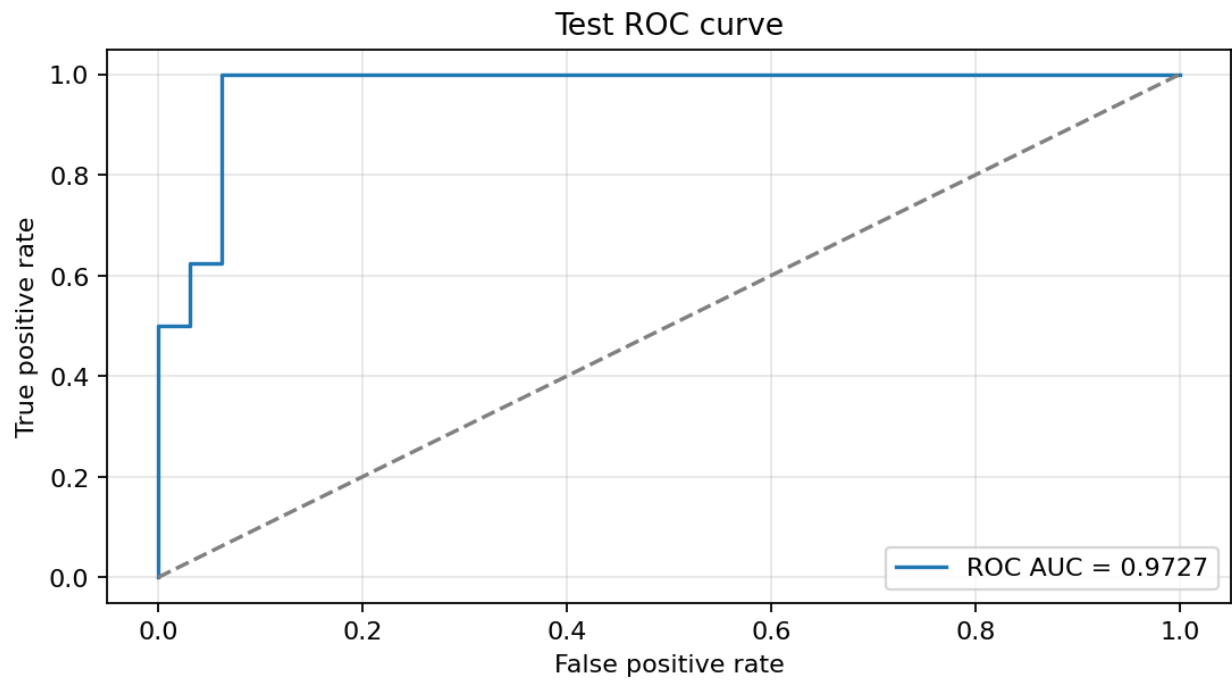
Dataset Distribution



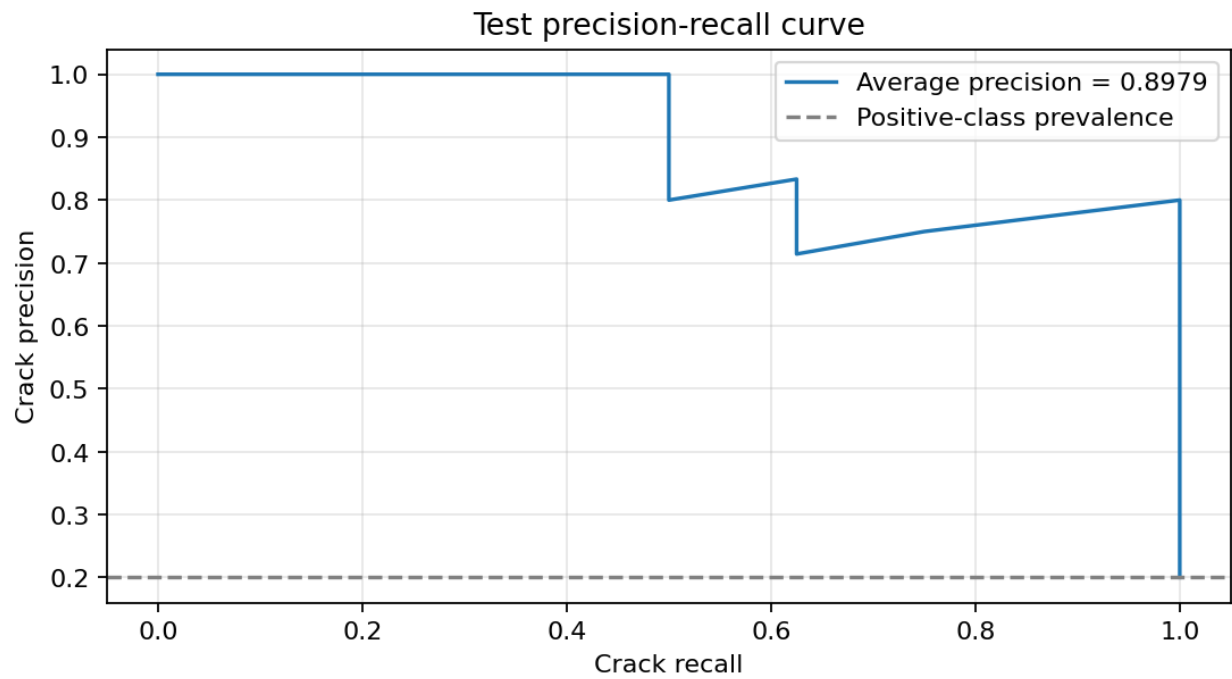
Test Confusion Matrix



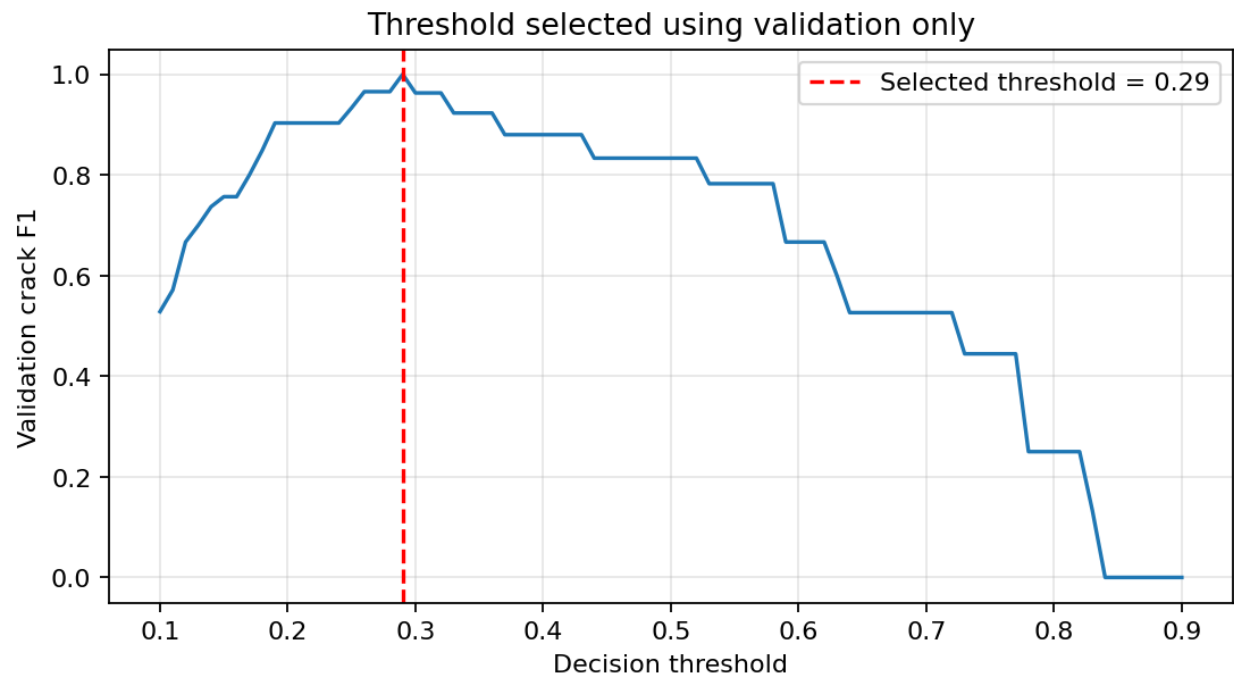
Test Roc



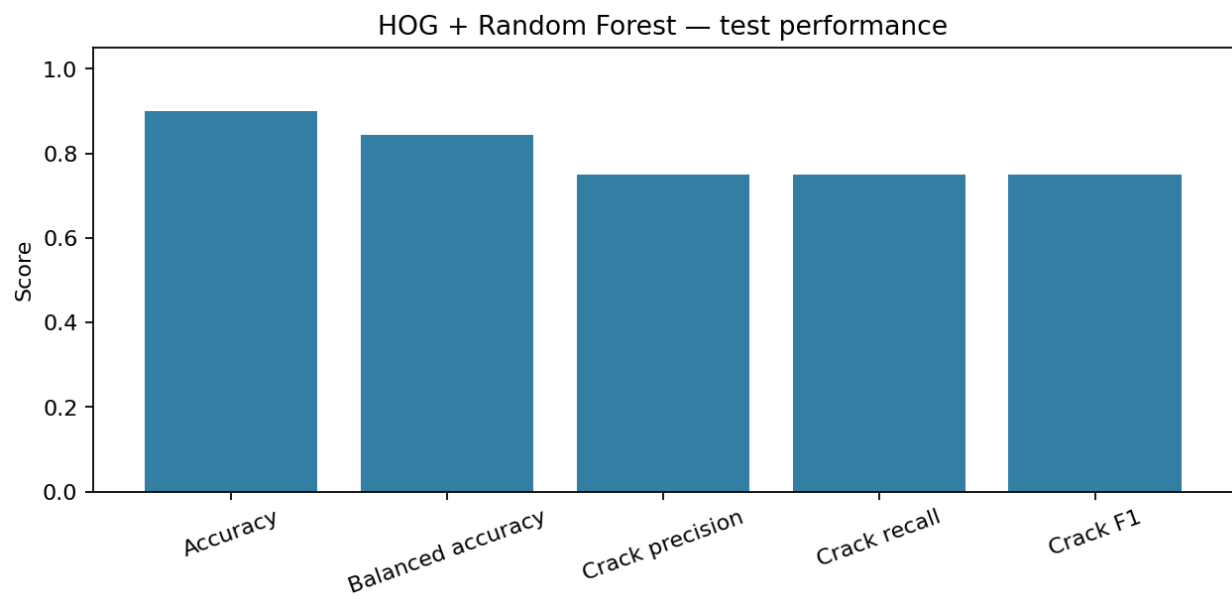
Test Precision Recall



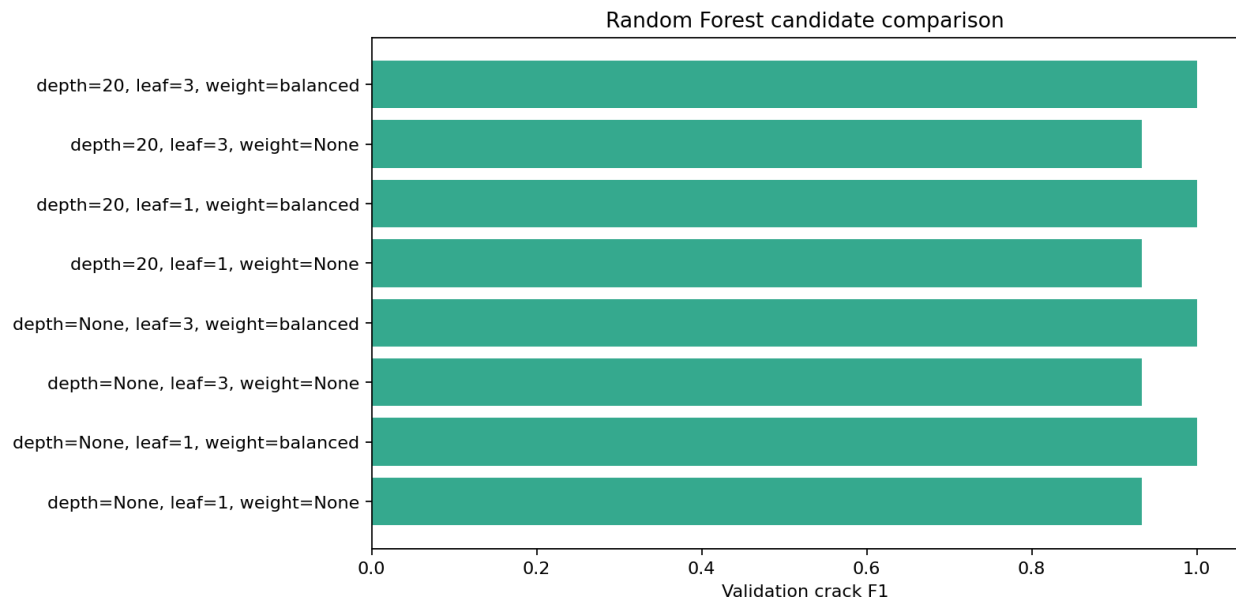
Validation Threshold



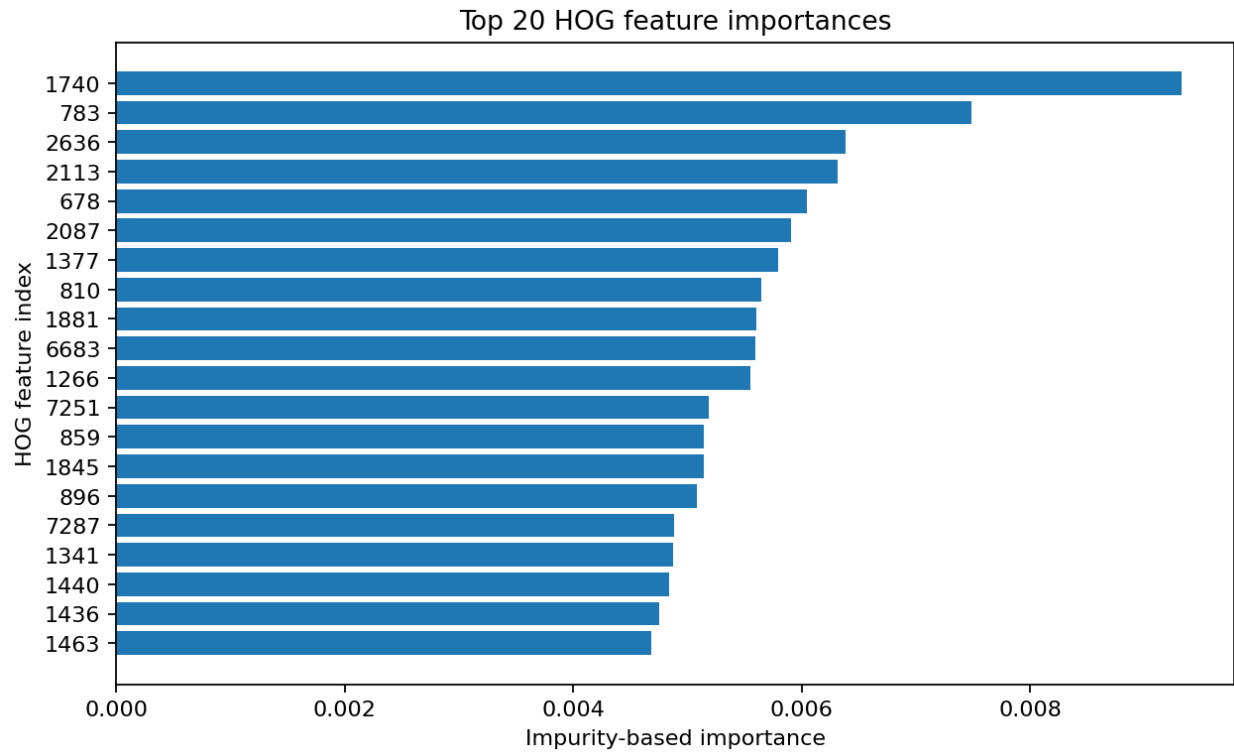
Test Metrics



Validation Search



Feature Importances



Hog Example

Preprocessed training patch



HOG visualization

